

Glycogen Turnover Forms Lactate during Exercise

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SHULMAN, R.G. Glycogen turnover forms lactate during exercise. *Exerc. Sport Sci. Rev.*, Vol. 33, No. 4, pp. 157–162, 2005. *The power needs of muscle are supplied by rapid anaerobic glycogenolytic ATP generation. Lactate is built up by the mismatch between steady state energy needs and short-term power demands and the increased concentration drives the lactate shuttle. In this model, contributions to fatigue should be looked for in the flux of glucose through glycogen rather than in the concentrations of fuel.*
Key Words: glycogenolysis, fatigue, exercise creatine kinase, lactate shuttle, skeletal muscle, acidosis

INTRODUCTION

The modern age of lactate studies began approximately 20 yrs ago when George Brooks questioned the accepted findings of previous generations and proposed the Lactate Shuttle (1). Early studies demonstrated that lactate accumulated when frog muscles contracted up to the point of exhaustion but disappeared during recovery in the presence of oxygen. In that view, the creation of lactate by exercising muscle is caused by the deficit of oxygen. However, numerous experiments by Brooks (1) and others have shown that lactate is generated during the performance of work by skeletal muscle in the presence of plentiful levels of oxygen (10). In fact, it is generated even when mitochondria are fully oxidized (10). NMR studies of the oxygenation of muscle myoglobin in exercising humans showed that oxygen levels in working muscle decrease with load, but even at maximal oxygen consumption are well above the mitochondrial needs (8).

The shuttling of lactate to redistribute energy led to questions about its origin, which was long assumed merely to be the result of an inadequate oxygen supply. This review is based on the understanding that lactate can shuttle energy from locations where it is synthesized, such as white fibers in skeletal muscle or astroglia in brain, to other locations, such as red muscle fibers or synaptic neurons, where it can be oxidized (1). The residual reductive capacity of lactate to

provide energy is not wasted by the body and its formation is not caused by a limited supply of oxygen. Lactate is not simply an unwanted byproduct, but rather is purposefully synthesized during work to meet normal physiological needs. However, the nature of that normal physiological mechanism has not been found. Now that the hypothesis postulating a deficiency of oxygen has been discarded, the reason for lactate generation to meet the energy needs of muscle and the brain requires an explanation. In this review, results from several avenues of research on the energetics of skeletal muscle are coordinated to propose a model for lactate buildup during muscle work under well-oxygenated conditions, a buildup that drives the Lactate Shuttle.

Evidence for Rapid Energy Consumption in Muscle

The energetic mechanisms that have been proposed for muscle reflect experimental measurements of fuels and oxygen consumption made in laboratory times of seconds to hours. However, we know that neuronal spiking and muscle contractions occur in milliseconds, and now it is known (2) that energy is supplied on this time scale in support of these rapid processes. During moderate-intensity exercise, sequential contractions will generally be separated by intervals that are orders of magnitude longer than the contractile time. Just how rapidly energy is obtained from PCr during single contractions was shown directly by the brilliant experiments of Chung *et al.* (2). Reasoning that even the fastest freeze-clamp measurements, with a time resolution of ~100 ms, did not follow high-energy phosphates during contraction and relaxation, they developed a gated ^{31}P NMR technique with ~1 ms time resolution to measure phosphate metabolites in the rat gastrocnemius muscle during 1-Hz stimulation. In this

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experiment, the NMR pulse-acquisition sequence was triggered synchronously with muscle stimulation and the spectra, acquired in several hundred scans, were summed. ATP remained constant at all times during contractions, whereas PCr rapidly decreased by $3 \mu\text{mol}\cdot\text{g}^{-1}$ tissue per twitch with a half-time of 8 ms. PCr then recovered with a half-time of 14 ms so that the whole PCr response to a twitch was finished in ~ 30 ms, too rapidly to have been observed by even the fastest extraction methods (Fig. 1). Chung *et al.* showed that this consumption of PCr/twitch is ~ 40 times greater than the values reported by dividing the decline in PCr after several minutes of contraction by the number of twitches. This ^{31}P NMR study showed that “the traditional method of calculating PCr/twitch underestimates the high-energy consumption arising from the drop and subsequent restora-

tion of the PCr pool during the millisecond contraction cycles” (2).

Chung *et al.*'s experiments show that PCr cannot be the ultimate source of energy in contracting muscle. At a cost of 3 mmol PCr/twitch, the muscle would be rapidly depleted of energy unless PCr were being replenished. The experiments show that PCr recovered between contractions. The question then becomes not whether PCr supports the contractions, but what source of energy replenishes the PCr levels after a contraction.

Interplay Between Glycogen, PCr, and Lactate during and after a Muscle Contraction

To understand how these rapid energetic processes are fueled, we must consider the rate of energy delivery by the available sources. ATP is produced much faster anaerobically (*i.e.*, by glycolysis or glycogenolysis), which converts glucose to lactate, than occurs by oxidative pathways, in which lactate is ultimately oxidized to CO_2 . The rapid resynthesis of PCr from ATP during a contractile period must utilize non-oxidative means—in skeletal muscle this review proposes that the ATP is produced by glycogenolysis. In working muscle, the ATP synthesized anaerobically could not come from glycolytic intermediates because the combined basal concentration of intramuscular glucose and glucose-6-phosphate is less than 0.3 mmol, which is insufficient to provide enough flux even if completely depleted (12). Furthermore, glucose transport flux, which is effectively unidirectional and therefore equal to the average net uptake measured during exercise, requires recruitment and is presumably too slow to support the millisecond energy flux. We suggest that glycogenolytic ATP production supplies the energy for millisecond bursts. This hypothesis implies that $\sim 1 \mu\text{mol}\cdot\text{g}^{-1}$ tissue of glycogen subunits is consumed during each contraction to refill the ATP and PCr pools. Because basal glycogen concentrations of $\sim 70 \mu\text{mol}\cdot\text{g}^{-1}$ tissue are not depleted even after several dozen contractions, resynthesis of glycogen must occur between twitches. Without the millisecond time pressure of contractions, the ATP required for this resynthesis can be achieved via somewhat slower oxidative means.

Although oxidation cannot supply ATP in milliseconds, it could resynthesize glycogen in the ~ 1 -s interval between even rapid contractions. Measured rates of oxygen consumption vary, seemingly dependent on the measurement method. None of the reported values reaches the rate of $1 \mu\text{mol}\cdot\text{g}^{-1}$ tissue per twitch, but the fastest values measured do reach several $\mu\text{mol}\cdot\text{g}^{-1}$ tissue per second. Hence, it is possible that the glycogen pools, which are decreased to refill PCr and ATP in milliseconds, are replenished by the energy supplied oxidatively in the longer period, no shorter than ~ 1 s, between contractions. In this model, glycogen decreases non-oxidatively in milliseconds to refill the ATP/PCr pool while it will be resynthesized oxidatively in periods of ~ 1 s between contractions.

Between contractions, a fraction of the lactate is oxidized to restore PCr and glycogen stores. However, the appearance of lactate during exercise shows that not all the lactate generated is oxidized between contractions. The faster the

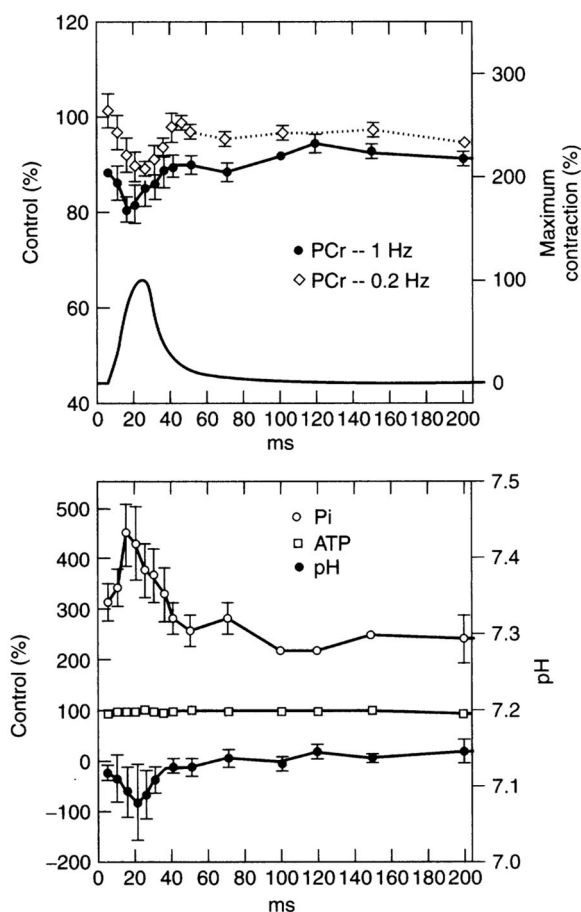


Figure 1. Metabolic transients during a twitch. The level of PCr declines by approximately 10% with a minimum at 16 ms and its time course is a slightly time-shifted mirror image of the force response, which exhibits a maximum at 25 ms. A similar PCr profile is observed under 0.2 Hz stimulation (. . .). The P_i increases approximately 1.8-fold and is inversely related to the PCr response. Based on the reported resting state values of PCr and P_i of 27.1 and $2.8 \mu\text{mol/g}$ wet tissue in rat gastrocnemius muscle, the calculated changes in PCr and P_i reaction appears to be stoichiometric. Concomitantly, pH appears to shift by 0.06 pH units, from 7.14 ± 0.11 . However, the limited accuracy of the measurement does not indicate a statistically significant change and precludes any definitive interpretation at this time. Corresponding P_i and pH profiles indicate a stoichiometric increase in P_i during the contraction cycle. (Reprinted from *Metabolomics by In Vivo NMR*, edited by R.G. Shulman and D.L. Rothman. West Sussex, England: John Wiley and Sons, Ltd., 2005, p. 115. Copyright © 2005 John Wiley and Sons, Ltd. Used with permission.)

contraction, the less lactate will be oxidized in the decreasing time between contractions. Some accumulated lactate is transported out of the muscle into the blood and thence to other locations, where it can be eventually oxidized or converted in the liver to glucose. Because transport is concentration-dependent, the concentration of lactate in a regularly contracting muscle will increase until a steady-state balance is reached between transport out of the muscle cell and net lactate production via this mechanism. Hence, the driving force for the lactate shuttle is its high concentration, created by the need for rapid nonoxidative energy production that exceeds the rate of lactate oxidation.

A common explanation for lactate build-up is that during exercise lactate/pyruvate are produced faster than the oxidative pathways can respond. This explanation postulates a limiting maximum rate for the TCA and ox-phos pathways, but this has not been established. It seems counterintuitive that the pathway of glucose oxidation, where flux is controlled by the demand for ATP, should falter under normal work conditions. However the explanation being offered here is that the inefficient glycogen shuttle, which produces twice as much lactate/rapid ATP as glycolysis, generates extra lactate while producing the same rapid ATP/contraction. During rapid contractions that cannot be supported by glycolysis, inefficient glycogenolysis produces two extra lactates per glucose consumed (12,13). Coyle has shown, for example, that the total rate of consumption of glucose as free glucose plus glycogen moieties is constant during several hours of exercise (3,4) and that fatigue can be delayed by the high rate of glucose infusion needed to maintain constant blood glucose. They found that glycogenolysis is the exclusive producer of glucose moieties at first, but as glycogen concentrations level off the uptake of glucose replaces the consumption of glycogen. The glycogen-shunt hypothesis has glycogen producing ATP and lactate rapidly by anaerobic metabolism during the entire exercising period with net glycogenesis increasing until the concentration of glycogen reaches a dynamic steady state. Evidence for this hypothesis from ^{13}C NMR experiments is presented.

Measurements made in muscle or brain are often made in times of seconds to hours. At these times, it is known that stimulation results in more glucose or glycogen being consumed than expected from the oxygen consumption, leading to the conclusion that the work is supported by the inefficient anaerobic production of energy; this is the oxygen deficit in muscle and the decrease in oxygen to glucose uptake in brain. However, the inefficiency is not as severe as has been thought, because the lactate is shuttled elsewhere either to be oxidized or reconverted to glucose. When glycogen is used as the source of lactate, as proposed, glycogen, PCr, and lactate are all partially to fully restored during the intervals between contractions when most of the incremental oxidative ATP production takes place.

Evidence That Glycogen Rather Than PCr Generates ATP during Contraction

The conventional view of short-term muscle energetics is that PCr supplies almost all of the energy needed for a sustained burst of contractions lasting less than 10 s, after

which it is replaced by glycogenolysis. This view is not supported by experiments. In a recent review, Greenhaff and Timmons (5) report, "It is now accepted, however, that PCr hydrolysis and lactate production do not occur in isolation, and that both are initiated rather rapidly at the onset of contraction."

The proposal that glycogenolysis is the ultimate energy source for exercise is supported by studies on patients with McArdle's disease. In this genetic metabolic disorder, glycogen phosphorylase is completely inactive. Patients experienced exercise intolerance and muscle cramping. ^{31}P NMR studies (11) of the resting muscle in these patients showed normal PCr, ATP, and Pi levels, but a decidedly higher pH (~ 7.2) than in control subjects (7.02 ± 0.01). During either moderate or high-intensity short-term exercise, PCr decreased much more rapidly in the patients than in the control subjects, whereas the pH remained high, and ATP levels remained constant. The inferred minimal lactate production in the McArdle's patients relative to control subjects is evidence that normal muscle uses glycogen to satisfy its contractile energy requirements. Furthermore, the normal levels of PCr present in these patients did not supply enough energy to support mild exercise in the absence of glycogenolytic flux.

Additional evidence that PCr is not the exclusive direct energy-supplying pathway comes from genetic knockout experiments. Watchko *et al.* (15) created mutant mouse strains missing either or both forms of creatine kinase and compared functional diaphragm muscle performance with control animals during isotonic activation. Both singly mutated forms showed no difference from control animals in force, velocity, power, time of sustained shortening, and work output, whereas these were reduced in the doubly mutated animal CK(−/−): a 20% decrease in the velocity of shortening, 16% decline in maximal power, 30% reduction in work, and a 40% attenuation in the time to sustain shortening during repetitive isotonic activation. However, in contrast to these moderate functional decreases, the creatine kinase activity in CK(−/−) mutants had decreased to less than 1% of control values. These results showed that PCr did not serve as a steady-state source of energy. With only 1% of creatine kinase activity, muscle force was only partially ($\sim 20\%$) reduced (14). These experiments force us to discard the notion that PCr provides the entire short-term energy for a contraction. Although there is a role for PCr in providing the proximate energy for contraction by buffering ATP, the temporal coordination of oxidative and nonoxidative pathways supply the ultimate energy for contraction. ATP would not be well buffered in the CK(−/−) mutants, which presumably accounts for the partial reductions in force.

Millisecond Coordination of Work and Energy by Ca^{2+}

The proposed model of rapid energy consumption by contraction and energy production by glycogenolysis is consistent with simultaneous Ca^{2+} triggering of the separate rates.

In addition to activation of the crossbridge cycle, the release of Ca^{2+} from the sarcoplasmic reticulum also triggers glycogenolysis, thereby supporting the hypothesis that

glycogen serves as the fuel for the millisecond energetics. Glycogen phosphorylase (GP) controls the rate of glycogenolysis. Phosphorylated GP is the active form. GP is phosphorylated by the enzyme phosphorylase kinase (PK), which in turn is activated by cAMP-dependent protein kinase (cAPK). Hormones maintain an adequate concentration of cAMP during exercise so that cAPK creates the activated conformation of PK. However, the Ca^{2+} concentration must increase for this conformation of PK to be activated. Hence, PK activity is controlled by Ca^{2+} during exercise, which increases in milliseconds and returns to basal levels. Hence, contraction and the rate of glycogenolysis are controlled by Ca^{2+} concentrations in the millisecond regime.

Evidence from ^{13}C NMR of Human Muscle for Glycogen Turnover during Exercise

There are many studies of the metabolic roles of muscle glycogen in exercise. High glycogen levels improve endurance (4,5), whereas depletion of glycogen is often associated with the onset of exhaustion. However, the specific need for glycogen in long-term moderate exercise has not been explained, and no specific connection between glycogen levels and fatigue has been possible. Nor have the steady-state measurements of glycogen concentrations been related to rapid energy bursts during the millisecond contractions of muscle fibers. Holloszy and Kohrt (7), in summarizing fundamental questions that remain about muscle energetics, asked “Why is muscle glycogen necessary for exercise of moderate and high intensities?”

Our suggested answer to this question is that glycogenolysis supplies the ATP needed within milliseconds of contraction and that the glycogen pools are replenished between contractions. A consequence is that during exercise glycogen is continually synthesized and degraded so that plasma glucose flows in and out of glycogen. ^{13}C NMR experiments of glycogen during exercise have supported this proposition. The glycogen C-1 resonance is a narrow, well-resolved line in the ^{13}C NMR spectrum. In muscle and liver, the 1.1% natural abundance ^{13}C NMR peak has good signal strength (Fig. 2). Comparison with calibrated solutions and with muscle biopsy data show that the ^{13}C peak intensity comes from all of the glucose moieties in high-molecular-weight glycogen (14). Thus, the intensity can be used to measure the concentration of glycogen. Because the accuracy of this determination is several times better than that provided by biopsies (14), it has been possible to extend some of the earlier studies.

One relevant illustration of the improvement in sensitivity over earlier work is seen in Figure 3, which plots glycogen concentrations in human gastrocnemius during medium intensity exercises with the plantarflexor muscles over several hours (9). Although earlier biopsy measurements had suggested that glycogen reached constant levels during this exercise, the high degree of constancy and its dependence on exercise intensity in individuals were certified by the NMR results. The level of glycogen reached lower levels with increasing intensity. This decline in glycogen depended on the load and not the previous history. Recovery was biphasic

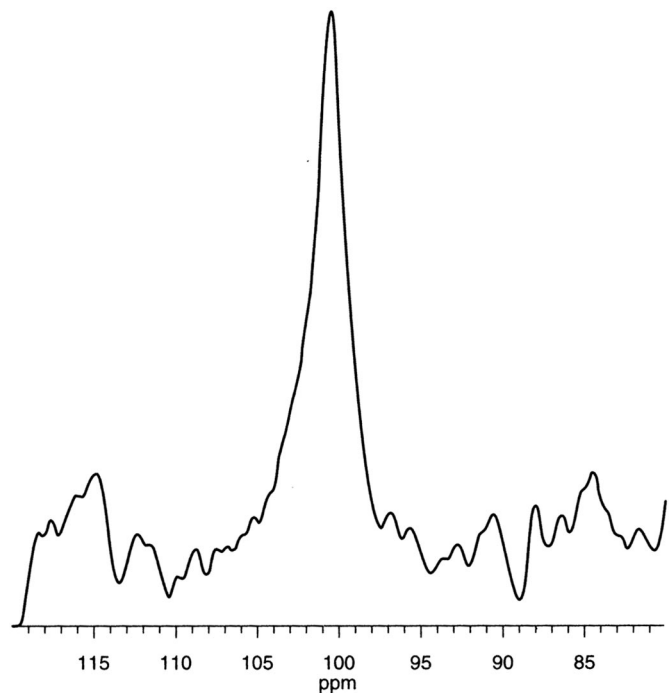


Figure 2. Glycogen C-1 resonance from human gastrocnemius at 100.5 ppm obtained by natural abundance ^{13}C nuclear magnetic resonance spectroscopy at 4.7T. The 5-min scan was performed while the subject was at rest. The signal-to-noise ratio was 21:1. (Reprinted from Price, T.B., D.L. Rothman, M.J. Avison, P. Buonamico, and R.G. Shulman. ^{13}C NMR measurements of muscle glycogen during low-intensity exercise. *J. Appl. Physiol.* 70:1836–1844, 1991. Copyright © 1991 The American Physiological Society. Used with permission.)

during an exercise course of intense toe lifts that resulted in a ~60% depletion in glycogen. The initial recovery was rapid and independent of insulin, whereas the remainder of the recovery to basal levels was slower and insulin-dependent (data not shown). However, when medium intensity exercise of the same muscle was conducted after the intense depletion, glycogen levels did not return to the preexercise baseline, but to the value characteristic of the plantar flexion exercise (Fig. 4). The ability of the muscle either to synthesize or deplete the glycogen stores highlighted the question: does the constant level of glycogen observed reflect a balanced turnover of synthesis and degradation or simply an inert pool? This question was answered by observing the rate at which labeled 1- ^{13}C glucose was incorporated into the glycogen pool (Fig. 5). Thus, during the several minutes needed for the NMR measurement the glycogen was in a dynamic state of synthesis that involved glycogen synthesis and degradation.

At the same time as these ^{13}C NMR studies were being performed, Hardin and Kushmerick (6) independently observed turnover of ^{13}C glucose moieties in muscle glycogen by ^{13}C NMR and showed the dynamic nature of the glycogen pool. In one experiment, they first labeled glycogen in segments of carotid artery smooth muscle with 2- ^{13}C glucose and then contracted the muscles in the presence of 1- ^{13}C glucose. In the presence of NaCN, oxidation was prohibited and significant signals from ^{13}C labeled lactate were observed. The time course showed that as the 2- ^{13}C -labeled

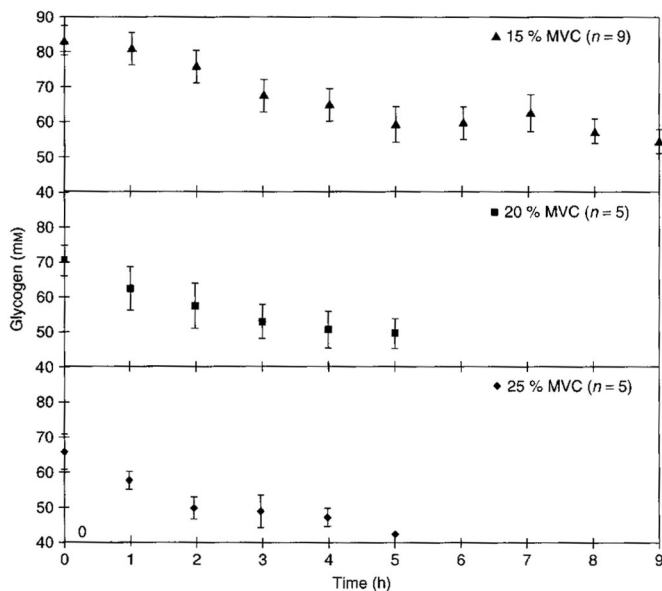


Figure 3. Glycogen concentration in gastrocnemius during low-intensity exercise at 15, 20, and 25% of maximal voluntary contraction (MVC). Values are means $\pm$ SE (Reproduced with permission. (Reprinted from Price, T.B., D.L. Rothman, M.J. Avison, P. Buonamico, and R.G. Shulman. ¹³C NMR measurements of muscle glycogen during low-intensity exercise. *J. Appl. Physiol.* 70:1836–1844, 1991. Copyright © 1991 The American Physiological Society. Used with permission.)

glycogen peak decreased over the first hour, the corresponding 2-¹³C lactate peak at first increased and then leveled off, showing the effects of glycogen breakdown. Toward the end of the first hour, the 3-¹³C lactate peak, which was derived from 1-¹³C glucose, appeared and increased during the next hour of the experiment. These results complement our ¹³C glycogen studies, during the first hour of which we saw breakdown of glycogen and in the subsequent period of constant glycogen a turnover from labeled glucose. This mechanism would not require separate pathways or compart-

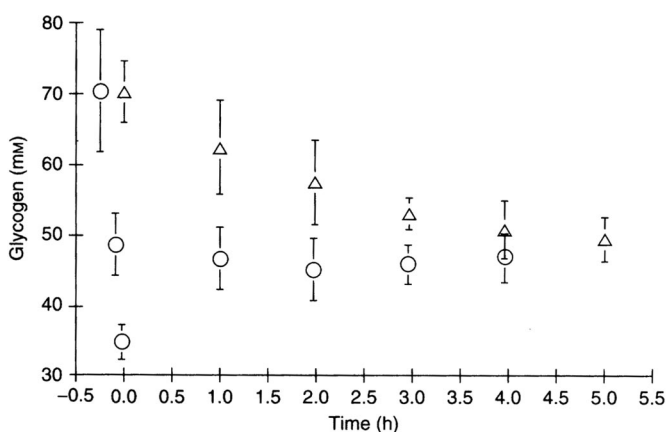


Figure 4. Glycogen concentration in human gastrocnemius during low-intensity exercise (20% MVC) from rest (Δ , $N = 5$) and for low-intensity exercise (20% MVC) after predepletion exercise ($\circ$, $N = 5$). Values are mean $\pm$ SE. (Reprinted from Price, T.B., D.L. Rothman, M.J. Avison, P. Buonamico, and R.G. Shulman. ¹³C NMR measurements of muscle glycogen during low-intensity exercise. *J. Appl. Physiol.* 70:1836–1844, 1991. Copyright © 1991 The American Physiological Society. Used with permission.)

ments for glycolytic and glycogenolytic fluxes, as had been proposed (6).

Possible Role for Glycogen during Fatigue

A long-standing question in studies of exercise physiology is why, under conditions in which there is adequate fuel from other sources (glucose, fatty acids) to sustain function, the reduction of glycogen leads to exhaustion. The model proposed and results described here may provide a solution to this paradox. Glycogen consumption has traditionally been linked to fatigue as the source of excess lactate and associated acidification, whereas glycogen reduction has been equated with an inability to provide sufficient energy to continue work. However, these explanations are incomplete. In fatigued McArdle patients, phosphorylase deficiency results in higher pH than in resting control subjects, and many other experiments have dissociated lactate generation and acidosis from muscle fatigue. Although it may play a role, particularly in intense anaerobic exercise, lactate acidosis is not the exclusive cause of fatigue.

Normal subjects performing moderately heavy exercise will experience fatigue at low levels of glycogen—even though the constant glycogen levels show that the net usage of glycogen for energy is insignificant (Fig. 3). In one study, glycogen levels were measured during exercise with and without continuous ingestion of glucose (3,4). Glycogen decreased at approximately the same rate in both cohorts, but the fed subjects took 1 h longer to reach exhaustion. In both cases, exhaustion occurred at a finite, apparently constant, concentration of glycogen, when glucose consumption had almost completely replaced net glycogen consumption as an energy source. The paradoxical finding that the net energy supplied by glycogen at exhaustion is small, despite the appearance of fatigue at low glycogen levels, could be explained by the glycogen-shunt model. The energy being provided for the work is not coming from glycogen, but through glycogen from plasma glucose. The energetic component of fatigue may be the result of an inability to provide the power requirements of the contraction through glycogenolysis. In this proposed explanation, glycogen does not serve as an energy source during prolonged exercise, because it remains constant. However, it does serve as a step in the pathway with a high rate of energy delivery or power. A decrease in this flux would not allow the muscle to perform the required work and cause fatigue. The rate of glucose flow through glycogen rather than glycogen concentrations should be explored as a factor that limits the time to exhaustion.

Summary of the Glycogen Shunt and Lactate Generation

1. Triggered ³¹P NMR shows that muscle contraction requires rapid, nonoxidative ATP production in milliseconds.
2. Glycogenolysis and muscle contractions are both switched on in milliseconds by Ca²⁺.
3. Glycogenolysis supplies the ATP needed during the milliseconds of contractions to replenish PCr.

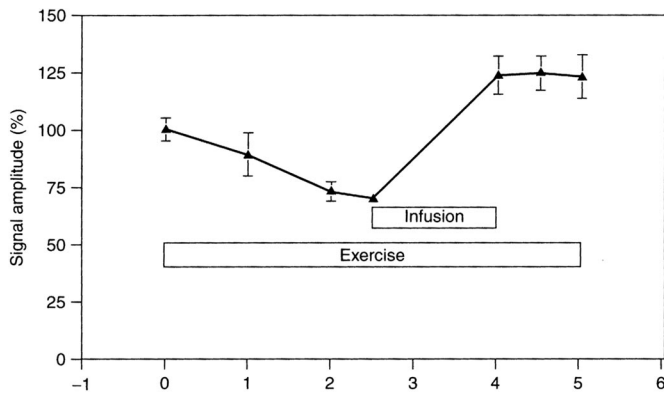


Figure 5. Change in gastrocnemius glycogen $1\text{-}^{13}\text{C}$ signal amplitude with exercise before and after infusion of 99% enriched $1\text{-}^{13}\text{C}$ glucose. Values are means $\pm$ SE for five subjects. Boxes denote exercise and infusion periods. After an infusion, the mean signal amplitude rose to $123 \pm 7\%$ of mean resting amplitude and plasma enriched in $1\text{-}^{13}\text{C}$ glucose increased to $21 \pm 2\%$ above natural abundance. (Reprinted from *Metabolomics by In Vivo NMR*, edited by R.G. Shulman and D.L. Rothman. West Sussex, England: John Wiley and Sons, Ltd., 2005, p. 132. Copyright © 2005 John Wiley and Sons, Ltd. Used with permission.)

4. The glycogen pool is replenished between contractions and glycogen serves as a temporal energy buffer.

5. Lactate accumulates in the muscle cell until transport out of the cell equals net production. This accumulation reflects a mismatch between the inefficient rapid ATP production by glycogenolysis and the oxidative processes that provide most of the net energy production over a contraction cycle.

6. Fatigue occurs during heavy aerobic exercise when the glycogen concentrations are low, but relatively constant. In the proposed mechanism, muscle work is supported by the glucose flux through glycogen, so that decreases in flux rates rather than concentration influence the time to exhaustion.

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